

2. Smart drip irrigation systems using IoT: a review of architectures, machine learning models, and emerging trends

This review paper explores the role of **IoT-based smart drip irrigation systems** in modern agriculture. The integration of **IoT, AI, and Machine Learning** enables real-time monitoring and automated irrigation management. Sensors collect data on **soil moisture, temperature, humidity, and weather conditions**, which helps optimize water distribution. The study reviewed **56 research articles published between 2015 and 2024** to analyze recent developments in this field. Machine learning models such as **Random Forest, ANN, KNN, and SVM** were used to improve irrigation decisions, with some studies reporting **up to 99.8% accuracy**. Smart irrigation systems have demonstrated **37–70% water savings** while enhancing crop productivity and reducing labor requirements. The use of **solar energy and cloud platforms** further increases sustainability and remote accessibility. However, challenges such as **high installation costs, poor rural connectivity, cybersecurity risks, and limited technical knowledge among farmers** remain barriers to widespread adoption. The review also highlights future trends, including **5G networks, AgroBots, edge computing, and Industry 5.0 applications** in agriculture. Overall, IoT-driven drip irrigation offers a promising approach to achieving **sustainable farming, efficient resource management, and improved food security**.